

Supporting Information

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SI Text

Proof of Minimal Spin-Orbit Coupled Filling in the Matrix-Product-State Formalism

In the matrix-product-state formalism, a short-range correlated translation invariant wave function Ψ is written as

$$\Psi_{\{a_n\}} = \sum_{\{\alpha_n\}=1}^{\chi} \prod_n B_{\alpha_n \alpha_{n+1}}^{a_n}, \quad [\text{S1}]$$

where the local Hilbert spaces are indexed by a_n and the α_n are “auxiliary” indexes to be summed over. For our purposes, χ can actually be infinite: What is important is that the Schmidt decomposition has a well-behaved and discrete spectrum, which follows from the area law obeyed by gapped ground states (15). An important question is how the global symmetries of Ψ manifest themselves in the local tensors B . In our case the global $U(1)$ and time reversal symmetries factorize into their onsite representations,

$$\hat{G}_\theta = \prod_n e^{i\theta N_n} = \prod_n g_\theta^{(n)}, \quad [\text{S2}]$$

$$\hat{G}_T K = \prod_n g_T^{(n)} K, \quad [\text{S3}]$$

where n label sites and the g are onsite unitary operators. Assuming translation invariance, we drop the superscript n . Standard results (28) show that in the absence of long-range order, Ψ is symmetric if and only if

$$\sum_b g_{\theta ab} B_{ab}^b = e^{i\nu\theta} \sum_{\alpha', \beta'} U_{\theta \alpha \alpha'}^\dagger B_{\alpha' \beta'}^a U_{\theta \beta' \beta}, \quad [\text{S4}]$$

$$\sum_b g_{T ab} \bar{B}_{ab}^b = \sum_{\alpha', \beta'} U_{T \alpha \alpha'}^\dagger B_{\alpha' \beta'}^a U_{T \beta' \beta}, \quad [\text{S5}]$$

for some unitary $\chi \times \chi$ matrices U , and $\nu \in \mathbb{Z}$ is the integer filling of the system.

The transformation laws of Eq. S5 define the U only up to a phase. Whereas the onsite symmetries satisfy

$$g_\pi = g_T g_T^*, \quad [\text{S6}]$$

because $T^2 = (-1)^{N_f}$, the U may realize the symmetry group only projectively,

$$U_T U_T^* = \omega(T, T) U_\pi, \quad \omega \in U(1). \quad [\text{S7}]$$

In our case this particular relation is not interesting, as we can choose the phase of U_π to set $\omega(T, T) = 1$. However, when we compare the equalities contained in Eqs. S5 and S6, we obtain

$$g_\pi B = e^{i\nu\pi} U_\pi^\dagger B U_\pi = g_T g_T^* B = U_T^T U_T^\dagger B U_T U_T^*, \quad [\text{S8}]$$

which implies

$$e^{i\nu\pi} B = B. \quad [\text{S9}]$$

Unless $\nu \in 2\mathbb{Z}$, we arrive at a contradiction.

On the Physical Principle Used in the Proof

Here we comment on our assumption of the nondegeneracy of a sym-SRE phase on flat, featureless geometries. The assumption was necessary because our results proved degeneracy only on an infinite sequence of geometries, and from this we wish to conclude that the phase cannot be sym-SRE in the thermodynamic limit. The situation is analogous to those of Oshikawa and Hastings, both of which bounded the gap only on odd by even tori. Regardless, their results are commonly interpreted to forbid sym-SRE paramagnets, implicitly invoking this principle. This principle is also implicit in many discussions of classification, so it would be useful to prove, but we do not attempt this here.

We motivate the principle more formally (but no more rigorously) in the case of a torus. An SRE phase is defined by the existence of a path of local Hamiltonians $\mathcal{H}_s, s \in [0, 1]$ from the Hamiltonian of interest $\mathcal{H}_0$ to a sum of single-site operators $\mathcal{H}_1$ while preserving the gap (12, 13). The interpolation $\mathcal{H}_s$ may break any of the onsite symmetries like time reversal; in fact, for symmetry-protected topological phases it must. However, because there are no examples of symmetry-protected topological phases protected only by translation, and because $\mathcal{H}_0$ is translationally invariant, it is reasonable to assume that the sequence of Hamiltonians can be chosen to preserve translation. Using translation invariance, the entire path $\mathcal{H}_s$ can be periodized and defined on a large torus, up to some exponentially small ambiguities if the terms in $\mathcal{H}_s$ are localized only exponentially. In the trivial limit the torus has a unique gapped ground state because $\mathcal{H}_1$ is a sum of single-site terms. Due to the finite gap on the plane, and hence short-range correlations, the periodized $\mathcal{H}_s$ should preserve the gap on the torus, in which case the ground state remains unique down to $s = 0$ —of course proving this is the main challenge.

There is a more restrictive setting where the argument is precise. Suppose that on the plane $\mathcal{H}_0$ can be exactly mapped to the trivial phase $\mathcal{H}_1$ via a finite-depth quantum circuit $U = \prod_a U_a$ with light cone of radius $R = \sum_a R_a$. We assume further that given some lengths $N_i > R$, the gates can be rearranged such that each stage U_a is invariant under the translations $T_{N_i}^{x_i}$. Because the U_a are composed of disjoint gates, for $N_i > R$ there is an obvious “furled” version U_{tor} defined on a torus of circumference N_i that has the same action as U on all local operators with a size smaller than $N_i - 2R$. Again, $\mathcal{H}_1$ can be trivially periodized onto the torus with a unique gapped ground state. Because U takes $\mathcal{H}_1$ to $\mathcal{H}_0$ on the plane, the finite light cone of U_{tor} also maps $\mathcal{H}_1$ to the periodized version of $\mathcal{H}_0$ on the torus. So they are isospectral, and $\mathcal{H}_0$ is gapped on the torus. More generally, the Bieberbach group will play the role of translation.

One drawback of the Bieberbach method is that it demonstrates exact degeneracy only on odd-volume spaces, whereas Hastings’ result rigorously bounds the scaling of the gap on an even by odd torus.

“Modding Out” and the Emergence of Spin and Pin $^\pm$ from a Lattice Model

Motivation and Setup. In *SI Text*, we clarify some subtleties in putting a system on a compact flat manifold $\mathbb{R}^D/\Gamma$. To illustrate the problem, let us first explore a more explicit form of the Hamiltonian on $\mathbb{R}^D/\Gamma$ obtained by the modding-out prescription explained in the main text.

To that end, let a subspace $M \subset \mathbb{R}^D$ be a fundamental domain that contains exactly one of the equivalent points of any $\vec{x} \in \mathbb{R}^D$. Namely, for any $\vec{y} \in \mathbb{R}^D$, there exist a unique $\vec{x} \in M$ and $g \in \Gamma$ such

that $\vec{y}=g(\vec{x})$. Note that g is also unique because Γ is fixed-point free. Although it is not necessary, for simplicity we choose M to be simply connected.

We assume the original Hamiltonian in $\mathbb{R}^D$ is a sum of local terms $\hat{H}=\sum_{\vec{x}}\hat{H}_{A_{\vec{x}}}$, where $\hat{H}_{A_{\vec{x}}}$ has a finite support $A_{\vec{x}}$ around $\vec{x}$. Then consider

$$\hat{H}_M = \sum_{\vec{x} \in M} \hat{H}_{A_{\vec{x}}}. \quad [\text{S10}]$$

When $\vec{x}$ in this sum is near the boundary of M , $\hat{H}_{A_{\vec{x}}}$ might contain operators outside of M ; in that case they should be understood as the identified operators inside M , given by the equivalence relation

$$\hat{O}_{\vec{x}}^{(i)} \sim_{\Gamma} \hat{g} \hat{O}_{\vec{x}}^{(i)} \hat{g}^{-1} = \hat{O}_{g(\vec{x})}^{(i)} (U_g)_{ji}. \quad [\text{S11}]$$

Then the natural identification $M \simeq M_{\Gamma} \equiv \mathbb{R}^D / \Gamma$ allows us to interpret $\hat{H}_M$ as the Hamiltonian on M_{Γ} . Note that the infinitely many different choices of M give the same M_{Γ} as long as one remembers each site in M_{Γ} can be labeled by the labels of any of the equivalent sites in $\mathbb{R}^D$.

Whereas the choice of g above, as a space group element, is unique, for SOC fermions the symmetry group has to be extended by the $\mathbb{Z}_2$ fermion parity and $\hat{g}$ is not uniquely defined. As such, a more careful discussion is warranted and we tackle it in the last subsection of *SI Text*.

Putting such complications aside, one still sees there are infinitely many valid choices of M . The choice of M , however, is merely a gauge choice. More precisely, in *SI Text* we show the following: Given any two valid choices of fundamental domains M and M' , the Hamiltonians $\hat{H}_M$ and $\hat{H}_{M'}$ are related by a local unitary transformation.

As an illustrative example, let us consider the following Hamiltonian in 2D,

$$\hat{H} = \sum_{l,m \in \mathbb{Z}} (-1)^l \hat{X}_{(l,m)} \hat{Y}_{(l+1,m)}, \quad [\text{S12}]$$

where X and Y are spin operators. Note that due to the $(-1)^l$ phase, the lattice translation in x is given by $l \mapsto l+2$. $\hat{H}$ is invariant under a space group $\mathcal{G}$ generated by glide G_x ,

$$\begin{aligned} G_x : (l, m) &\mapsto (l+1, -m), \\ \hat{G}_x \hat{X}_{(l,m)} \hat{G}_x^{-1} &= \hat{X}_{(l+1, -m)}, \\ \hat{G}_x \hat{Y}_{(l,m)} \hat{G}_x^{-1} &= -\hat{Y}_{(l+1, -m)}, \end{aligned} \quad [\text{S13}]$$

and translation T_y ,

$$\begin{aligned} T_y : (l, m) &\mapsto (l, m+1), \\ \hat{T}_y \hat{X}_{(l,m)} \hat{T}_y^{-1} &= \hat{X}_{(l, m+1)}, \\ \hat{T}_y \hat{Y}_{(l,m)} \hat{T}_y^{-1} &= \hat{Y}_{(l, m+1)}. \end{aligned} \quad [\text{S14}]$$

Let Γ be the subgroup of the full symmetry group generated by $G_x^{N_x}$ and $T_y^{N_y}$ with N_x, N_y being odd integers. As described, we introduce the equivalence relation

$$\hat{X}_{(l,m)} \sim_{\Gamma} \hat{X}_{(l, m+N_y)} \sim_{\Gamma} \hat{X}_{(l+N_x, -m)}, \quad [\text{S15}]$$

$$\hat{Y}_{(l,m)} \sim_{\Gamma} \hat{Y}_{(l, m+N_y)} \sim_{\Gamma} -\hat{Y}_{(l+N_x, -m)}. \quad [\text{S16}]$$

For example, M can be chosen as $\{(l, m) | 1 \leq l \leq N_x, 1 \leq m \leq N_y\}$. Under identification of the edges of M by Γ , one sees that M_{Γ} is actually the Klein bottle.

As described, the modded-out Hamiltonian is then given by

$$\begin{aligned} \hat{H}_M = \sum_{l=1}^{N_x-1} \sum_{m=1}^{N_y} (-1)^l \hat{X}_{(l,m)} \hat{Y}_{(l+1,m)} + \sum_{m=1}^{N_y-1} \hat{X}_{(N_x,m)} \hat{Y}_{(1, N_y-m)} \\ + \hat{X}_{(N_x, N_y)} \hat{Y}_{(1, N_y)}, \end{aligned} \quad [\text{S17}]$$

where we have used Eq. S16 to bring $\hat{Y}_{(N_x+1,m)}$ into the region M and $(-1)^{N_x+1} = 1$ as N_x is odd.

Staring at Eq. S17, one sees that

- Around the ‘‘seam’’ $l=1$, $\hat{H}_M$ contains couplings (boundary conditions) that do not seem to coincide with the original Hamiltonian defined on the plane.
- The symmetry of $\hat{H}_M$ is unclear—Does $\hat{H}_M$ still have something reminiscent of the $\mathcal{G}$ invariance of $\hat{H}$?

For the first point, we argue that the seam of M_{Γ} is an artifact of choosing a specific fundamental domain. We can see that the Hamiltonian (deep) inside M (the first sum in Eq. S17) is identical to that of the original Hamiltonian in $\mathbb{R}^2$. By redefining M , any point in $\mathbb{R}^2$ can be taken far away from the boundary of M . For example, for (l_0, m_0) ($l_0, m_0 \in \mathbb{Z}$), one may choose

$$M = \left\{ (l, m) \mid |l - l_0| < \frac{N_x}{2}, |m - m_0| < \frac{N_y}{2} \right\}. \quad [\text{S18}]$$

Therefore, one can argue that any expectation values of local operators or correlation functions around any given points in M_{Γ} are unchanged from those in the bulk (for sufficiently large N_x, N_y). This point can be made clearer, using the local unitary equivalence we discuss in the next subsection.

The answer for the second point depends on the property of Γ . To understand this, one should ask, Does $[\vec{x}] = [\vec{y}]$ always imply $[g(\vec{x})] = [g(\vec{y})]$ for all $g \in \mathcal{G}$? It is easy to see that this is the case only when Γ is a normal subgroup of $\mathcal{G}$. When Γ is normal, the $\mathcal{G}$ invariance of $\hat{H}$ implies the $\mathcal{G}$ invariance of $\hat{H}_M$ (up to local unitaries). Even when Γ is not normal, however, $\hat{H}_M$ is still invariant under the normalizer of Γ in the full symmetry group (e.g., in the above example, the normalizer of Γ in $\mathcal{G}$ is the subgroup generated by G_x). For our purpose, the $\mathcal{G}$ invariance of $\hat{H}_M$ is not necessary—the properties that each point of M_{Γ} is locally identical to the bulk and that $\hat{H}_M$ remains TR symmetric are sufficient.

Local Unitary Equivalence. Here we explicitly show the asserted local unitary equivalence among $\{\hat{H}_M\}$ defined with different fundamental domains and discuss its physical consequence.

To this end, we first define a slightly more formal, but nonetheless equivalent, procedure to define the modded-out Hamiltonian in Eq. S10. This merely serves to introduce notations that make the local unitary equivalence manifest. By definition, for any site $\vec{y}$ in $\mathbb{R}^D$ there exist $\vec{x} \in M$ and $g_M^{\vec{y}} \in \Gamma$, both unique, such that $\vec{x} = g_M^{\vec{y}}(\vec{y})$. In $\mathbb{R}^D$, the action of $g_M^{\vec{y}}$ on any operator $\mathcal{O}_{\vec{y}}$ is represented by first sending $\vec{y} \rightarrow \vec{x}$ and then conjugating $\mathcal{O}_{\vec{x}}$ by a local unitary operator $\hat{s}_M^{\vec{y}}$. Note that $\hat{s}_M^{\vec{y}}$ acts nontrivially only on site $\vec{x} = g_M^{\vec{y}}(\vec{y})$. After modding out, however, the relabeling $\vec{y} \rightarrow \vec{x}$ becomes trivial because $\vec{x}, \vec{y}$ are merely different labels for the same site. The identification of operators, as discussed in Eq. S11, can then be defined via the map $\mu_M : \mathcal{O}_{\vec{y}} \mapsto \hat{s}_M^{\vec{y}} \mathcal{O}_{\vec{x}} \hat{s}_M^{\vec{y}\dagger}$.

μ_M can be promoted to a linear map acting on any local operators with support in region $A \subset \mathbb{R}^D$, provided any two distinct sites in A belong to different equivalence classes. More concretely, we simply have $\mu_M(\otimes_{\vec{y} \in A} \mathcal{O}_{\vec{y}}) = \hat{U}_M^A(\otimes_{\vec{y} \in A} \mathcal{O}_{\hat{s}_M^{\vec{y}}(\vec{y})}) \hat{U}_M^{A\dagger}$ with $\hat{U}_M^A = \otimes_{\vec{y} \in A} \hat{s}_M^{\vec{y}}$ being a local unitary operator. We can now rewrite the modded-out Hamiltonian in Eq. S10 in a more precise manner:

$$\hat{H}_M = \sum_{\vec{x} \in M} \mu_M(\hat{H}_{A_{\vec{x}}}). \quad [\text{S19}]$$

In particular, because the elements of Γ commute with time-reversal $\mathcal{T}$, $\hat{H}_M$ is time-reversal symmetric.

Taken at face value, the two Hamiltonians $\hat{H}_M$ and $\hat{H}_{M'}$ obtained by modding out with fundamental domains $M \neq M'$ are not identical. However, the choice of the fundamental domain is only a gauge choice and one can always construct a local unitary transform $\hat{U}$ such that $\hat{U}\hat{H}_M\hat{U}^\dagger = \hat{H}_{M'}$. In fact, one simply sees that $\hat{H}_{M'} = \mu_{M'}(\hat{H}_M)$ and therefore $\hat{U} = \hat{U}_{M'}^M$. (Note again one can freely switch between the equivalent labels for the same site.)

Such local unitary equivalence implies the ground state expectation value of any local operators with support $A_\Gamma \subset M_\Gamma$ is identical to that of the bulk with exponential accuracy. If A_Γ corresponds to a region A deep inside M , this is evident because the ground state is gapped and the restriction of $\hat{H}_M|_{A_\Gamma}$ is identical to $\hat{H}|_A$. If A_Γ corresponds to a region A' that crosses the border of M (or equivalently the corresponding region is disconnected in M), one can nonetheless choose a new fundamental domain M' to center A' . In essence, given a “measurement region” A_Γ , one can move any apparent seams arbitrarily far away from it, using $\hat{U}$ (bounded by the size of M_Γ), and hence deduce the measurement outcome is identical to the bulk up to exponential corrections. Important to the discussion is that $\hat{U}$ leaves the region A_Γ unchanged. This should be contrasted with a “visible” defect, say an impurity, which can be moved only by invoking a translation operator and will therefore shift also the measurement region.

Complications with Spinful Fermions: Spin and Pin. As explained in the main text, the “spin” degrees of freedom complicate the situation because the internal rotations or reflections introduce sign factors that depend on the fermion number $\hat{N}$. Here we clarify the mathematical structure of Spin or Pin structure on a compact flat manifold $M_\Gamma = \mathbb{R}^D/\Gamma$ in more detail.

Before discussing (s)pin, let us first examine the Bieberbach group Γ itself. The first Bieberbach theorem states that each $g \in \Gamma$ has a unique decomposition into a product of a (possibly fractional) translation T and an orthogonal transformation $B \in O(D)$, $g = TB$. For our purposes only a discrete subset $G \subset O(D)$ will appear, such as reflections and $2\pi/m$ rotations. To be fancy, we have the short exact sequence

$$1 \rightarrow \mathbb{Z}^d \rightarrow \Gamma \xrightarrow{r} G \rightarrow 1, \quad [\text{S20}]$$

where $\mathbb{Z}^d$ denotes the lattice translations in Γ and r projects onto the point group part of g ; i.e., $r(TB) = B$. The map r is the “holonomy representation” of M_Γ . For the Bieberbach manifold M_Γ , the fundamental group is $\pi^1(M_\Gamma) = \Gamma$; intuitively this is because a straight line connecting two points on $\mathbb{R}^D$ related by $g \in \Gamma$ becomes a closed curve in M_Γ . The map r encodes the fact that when traversing a path $g \in \pi^1(M_\Gamma)$, we return to the point having been “screwed” by the orthogonal transformation $r(g) \in G$.

In the presence of (s)pin degrees of freedom, G is realized as a $\mathbb{Z}_2$ extension encoding the sign structure, which we denote by G^F . For instance, for spin-1/2, the 2π rotation is $\hat{C}_{2\pi} = (-1)^{\hat{N}}$, where $\hat{N}$ denotes the fermion number. The screw and glide transformations then inherit these signs, and a 2_1 screw will square to a translation and the fermion parity, $\hat{S}^2 = (-1)^{\hat{N}}\hat{T}$. Formally,

$$1 \rightarrow \mathbb{Z}_2 \rightarrow G^F \xrightarrow{\lambda} G \rightarrow 1, \quad [\text{S21}]$$

where $\lambda[(-1)^{\hat{N}}] = 1$. The most familiar example of this sequence is probably the case of $G = \text{SO}(3)$ and $G^F = \text{Spin}(3) \simeq \text{SU}(2)$. The choice of G^F depends on the physical model, but for SOC spin-1/2

fermions G^F is a discrete subgroup of the ‘Pin $^-(D)$ ’ group, the unoriented counterpart of Spin(D). The “ $-$ ” superscript indicates that mirror reflections satisfy $R^2 = (-1)^{\hat{N}}$, because reflection is implemented on spins as $R_x = i\sigma^x$. Generally, we can consider both Pin $^+(D)$ and Pin $^-(D)$, depending on whether reflections square to $R^2 = (\pm 1)^{\hat{N}}$.

To consistently introduce the equivalence relation on spinful fermions required to mod out by Γ , we need a group homomorphism $\epsilon: \Gamma \rightarrow G^F$, which encodes how we choose the sign that can accompany each reflection and rotation. Namely, ϵ has to satisfy

$$\epsilon(g_1)\epsilon(g_2) = \epsilon(g_1g_2), \quad [\text{S22}]$$

such that the following diagram commutes:

$$\begin{array}{ccc} & G^F & \\ \nearrow_r & \downarrow & \\ \Gamma & \longrightarrow & G. \end{array} \quad [\text{S23}]$$

This is a “lift” of the holonomy representation to G^F . Depending on Γ , it may or may not be possible to find such an ϵ . In fact, ref. 22 shows that Eq. S23 is precisely the condition for the existence of a Spin/Pin structure on M_Γ . All orientable 3-manifolds admit spin structure and all nonorientable 3-manifolds admit Pin $^-$ structure. Hence, all 10 Bieberbach 3-manifolds (6 orientable and 4 nonorientable) admit Spin or Pin $^-$ structure, and such ϵ always exists.

As an illuminating example, let us discuss the space group no. 29. This group is generated by three symmetry elements

$$G_x: (x, y, z) \mapsto \left(x + \frac{a_x}{2}, -y, z\right), \quad [\text{S24}]$$

$$T_y: (x, y, z) \mapsto (x, y + a_y, z), \quad [\text{S25}]$$

$$S_z: (x, y, z) \mapsto \left(-x, -y, z + \frac{a_z}{2}\right), \quad [\text{S26}]$$

with relations

$$G_x^2 = T_x, \quad S_z^2 = T_z, \quad G_x S_z = T_x S_z G_x, \quad [\text{S27}]$$

where $T_{x,y,z}$ is translation by one unit. Because $r(G_x) = IC_{2x}$ (I is the inversion) and $r(S_z) = \hat{C}_{2z}$, we have ϵ up to signs $\xi_{x,y,z} = \pm 1$:

$$\epsilon(G_x) = \xi_x e^{i(\sigma_x/2)\pi} = \xi_x i\sigma_x, \quad [\text{S28}]$$

$$\epsilon(T_y) = \xi_y \sigma_0, \quad [\text{S29}]$$

$$\epsilon(S_z) = \xi_z e^{i(\sigma_z/2)\pi} = \xi_z i\sigma_z. \quad [\text{S30}]$$

For ϵ to be homomorphic, ϵ has to keep relations in Eq. S27. It is easy to see that once we fix $\epsilon(T_x) = \epsilon(T_z) = -1$, all relations are satisfied without specifying $\xi_{x,y,z} = \pm 1$. We thus have $2^3 = 8$ Pin $^-$ structures.

Curiously, this group, no. 29, is the single exception that does not admit Pin $^+$ structure among all Bieberbach groups. For Pin $^+$ we have to use $\epsilon(G_x) = \xi_x \sigma_x$ so that $\epsilon(G_x^2) = +1$, which also fixes $\epsilon(T_x) = +1$. Then, the third relation in Eq. S27 is violated. Therefore, we cannot put particles obeying Pin $^+$ on the manifold “first amphidicosm” obtained by modding out $\mathbb{R}^3$ by $\Gamma = \text{no. 29}$. This, however, does not concern us here because we are interested in spin-1/2 electrons.

Note that because we consider models with U(1) symmetry, it seems that we could redefine the reflections by an arbitrary U(1)

phase θ , $R_x \rightarrow e^{i\theta N} R_x$. Choosing then $\theta = \pi/2$ erases the distinction between Pin^+ and Pin^- . Formally this amounts to considering $U(1)$, rather than $\mathbb{Z}_2$, extensions of the space group, leading to $\text{Spin}_{\mathbb{C}}$ and $\text{Pin}_{\mathbb{C}}$ structures. However, time reversal narrows down this $U(1)$ ambiguity because the modding-out process must preserve TR symmetry. The space group elements $g \in \Gamma$ and T commute; the lift ε must as well, $\varepsilon(g)T = T\varepsilon(g)$; otherwise the seams of the region will break TR. For $S=1/2$ models, this amounts to requiring

$$\sigma_y \varepsilon(g) * \sigma_y = \varepsilon(g). \quad [\text{S31}]$$

So TR invariance reduces the $U(1)$ ambiguity back down to $\mathbb{Z}_2$, fixing either $\text{Pin}^\pm$. For example, $\varepsilon(G_x) = \xi_x i \sigma_x$ for Pin^- satisfies this, whereas $\varepsilon(G_x) = \xi_x \sigma_x$ for Pin^+ does not. In general, the relation $R^2(TRT^{-1}R^{-1}) = (\pm 1)^N$ is invariant under $U(1)$ redefinitions of R (because T conjugates phases) and determines which of $\text{Pin}^\pm$ to use.

Modding-Out Procedure for Space Group No. 24

Here we discuss an explicit example of the modding-out procedure described in the main text and demonstrate how this leads to the filling constraint $\nu \in 4\mathbb{Z}$ for any sym-SRE phase in space group no. 24. For brevity, in this section we use the notation SG_n to denote the space group indexed by the number n in ref. 18.

To this end, we first focus on SG_{19} , which differs from SG_{24} only in their Bravais lattices: SG_{19} is primitive orthorhombic whereas SG_{24} is body-centered orthorhombic. In particular, we make use of the fact that SG_{19} can be regarded as a subgroup of SG_{24} .

Aside from lattice translations, SG_{19} contains three orthogonal 2_1 screws of the form

$$S_1 : (x, y, z) \mapsto \left(x + \frac{a_1}{2}, -y + \frac{a_2}{2}, -z\right); \quad [\text{S32}]$$

$$S_2 : (x, y, z) \mapsto \left(-x, y + \frac{a_2}{2}, -z + \frac{a_3}{2}\right); \quad [\text{S33}]$$

$$S_3 : (x, y, z) \mapsto \left(-x + \frac{a_1}{2}, -y, z + \frac{a_3}{2}\right), \quad [\text{S34}]$$

where a_i is the lattice constant along $\hat{x}_i$. In the following we always measure “volumes” in units of $a_1 a_2 a_3$, the volume of the primitive unit cell of SG_{19} .

Note that $S_i^2 = T_i$, where T_i is the lattice translation along $\hat{x}_i$, and $S_1 S_2 S_3 = 1$, the group identity. SG_{19} is therefore generated by

$\{S_1, S_2\}$. In particular, observe that $\{\vec{0}, S_1 \vec{0}, S_2 \vec{0}, S_3 \vec{0}\}$ are four distinct points not related by lattice translations. Hence, the lattice composing of all of the symmetry-related points of $\vec{0}$ has four sites in each unit cell, and each site “occupies” a volume of $1/4$.

Suppose we mod out $\mathbb{R}^3$ by the entire SG_{19} . As such, the four points $\{\vec{0}, S_1 \vec{0}, S_2 \vec{0}, S_3 \vec{0}\}$ and all their lattice translates are identified. The resulting manifold $\mathbb{R}^3/\text{SG}_{19}$ therefore has the same volume as that occupied by $\vec{0}$ in the original lattice; i.e., the resulting manifold (known as the “didicosm”; Fig. S2) has a volume of $1/4$.

In our argument, however, it is crucial that the resulting manifold $\mathbb{R}^3/\Gamma$ is locally indistinguishable from $\mathbb{R}^3$. This implies the following conditions on the subgroup Γ : (i) Γ is fixed-point free, and (ii) $\mathbb{R}^3/\Gamma$ has to be arbitrarily large in all linear dimensions. As discussed above, using $\Gamma = \text{SG}_{19}$ fails to satisfy condition ii. Intuitively, this can be cured by choosing a “smaller” Γ . More precisely, we demand Γ be chosen such that the images of (say) the origin constitute a superlattice with a unit cell that is arbitrarily large with generic aspect ratios.

This can be accomplished by choosing Γ to be the subgroup generated by $\{\tilde{S}_1 = T_2^{N_2} S_1^{2N_1+1}, \tilde{S}_2 = T_3^{N_3} S_2^{2N_2+1}\}$, where N_i are integers that can be arbitrarily large. Note that $\tilde{S}_3 = (\tilde{S}_1 \tilde{S}_2)^{-1}$, and $\tilde{T}_i = \tilde{S}_i^2 = T_i^{2N_i+1}$. In fact, Γ is identical to SG_{19} but with the rescaled lattice constants $\tilde{a}_i = (2N_i + 1)a_i$. The linear dimensions of $\mathbb{R}^3/\Gamma$ are therefore $O(N_i a_i)$, as desired.

It follows from our previous discussion that the volume of $\mathbb{R}^3/\Gamma$ is $(2N_1 + 1)(2N_2 + 1)(2N_3 + 1)/4$, an odd-integer multiple of $1/4$. To avoid Kramers degeneracy the electron filling has to be $\nu \in 8\mathbb{Z}$ for any sym-SRE phase symmetric under SG_{19} . Being a subgroup of SG_{19} , Γ is also a subgroup of SG_{24} . Hence the same filling constraint applies to SG_{24} —except that the volume of the primitive unit cell of SG_{19} is twice as big as that of SG_{24} . As a result, the sym-SRE filling constraint is $\nu \in 4\mathbb{Z}$ for SG_{24} .

Group-Subgroup Relations and Elementary NS Space Groups

As explained in the main text, all 157 NS space groups contain at least one of the elementary NS space groups as a t subgroup. Fig. S1 illustrates which space group contains which elementary space groups, generated based on ref. 18. Elementary space groups are grouped into 10 Bieberbach manifolds (19).

Using this group-subgroup relation, one can deduce the constraint on filling ν for all nonelementary NS space groups based on the result for NS elementary space groups established in the main text. We list the results in Table S1.

No.*	ν	No.	ν	No.	ν	No.	ν	No.	ν	No.	ν	No.	ν
4	$4n$	39	$4n$	66	$4n$	100	$4n$	129	$4n$	165	$4n$	201	$4n$
7	$4n$	40	$4n$	67	$4n$	101	$4n$	130	$8n$	167	$4n$	203	$4n$
9	$4n$	41	$4n$	68	$4n$	102	$4n$	131	$4n$	169	$12n$	205	$8n$
11	$4n$	43	$4n$	70	$4n$	103	$4n$	132	$4n$	170	$12n$	206	$4n^+$
13	$4n$	45	$4n$	72	$4n$	104	$4n$	133	$4n^+$	171	$6n$	208	$4n$
14	$4n$	46	$4n$	73	$4n^+$	105	$4n$	134	$4n$	172	$6n$	210	$4n$
15	$4n$	48	$4n$	74	$4n$	106	$4n^+$	135	$4n^+$	173	$4n$	212	$8n$
17	$4n$	49	$4n$	76	$8n$	108	$4n$	136	$4n$	176	$4n$	213	$8n$
18	$4n$	50	$4n$	77	$4n$	109	$4n$	137	$4n$	178	$12n$	214	$4n$
19	$8n$	51	$4n$	78	$8n$	110	$4n^+$	138	$8n$	179	$12n$	218	$4n$
20	$4n$	52	$8n$	80	$4n$	112	$4n$	140	$4n$	180	$6n$	219	$4n$
24	$4n$	53	$4n$	84	$4n$	113	$4n$	141	$4n$	181	$6n$	220	$4n^+$
26	$4n$	54	$8n$	85	$4n$	114	$4n$	142	$4n^+$	182	$4n$	222	$4n$
27	$4n$	55	$4n$	86	$4n$	116	$4n$	144	$6n$	184	$4n$	223	$4n$
28	$4n$	56	$8n$	88	$4n$	117	$4n$	145	$6n$	185	$4n$	224	$4n$
29	$8n$	57	$8n$	90	$4n$	118	$4n$	151	$6n$	186	$4n$	226	$4n$
30	$4n$	58	$4n$	91	$8n$	120	$4n$	152	$6n$	188	$4n$	227	$4n$
31	$4n$	59	$4n$	92	$8n$	122	$4n$	153	$6n$	190	$4n$	228	$4n^+$
32	$4n$	60	$8n$	93	$4n$	124	$4n$	154	$6n$	192	$4n$	230	$4n^+$
33	$8n$	61	$8n$	94	$4n$	125	$4n$	158	$4n$	193	$4n$		
34	$4n$	62	$8n$	95	$8n$	126	$4n$	159	$4n$	194	$4n$		
36	$4n$	63	$4n$	96	$8n$	127	$4n$	161	$4n$	198	$8n$		
37	$4n$	64	$4n$	98	$4n$	128	$4n$	163	$4n$	199	$4n$		

$\nu = 8n - 4$ is prohibited for the noninteracting case. There is no known interacting model of sym-SRE at these fillings either.

$\nu=4$ is prohibited for the noninteracting case. There is no known interacting model of sym-SRE at this filling either.